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Original Article

Study of psychological effects of deployment at extreme high altitude area on soldiers

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ABSTRACT

Background: The Indian Armed Forces are fighting the battle at extreme High Altitude, the most inhospitable terrain in the world, for the last thirty five years. The stress of being isolated under harsh environmental conditions on a daily basis can have an adverse effect on their mind. However, so far, no study has been undertaken to assess the psychological effects of deployment at extreme High Altitude.

Methods: Three hundred thirty-four troops selected for deployment were initially evaluated using the General Health Questionnaire-12 (GHQ-12) and Armed Forces Medical College Life Events Scale (AFMC LES) as screening tools to assess mental health status after obtaining ethical clearance and informed consent. On deinduction after a deployment for more than three months, they were reassessed. The data collected were statistically analysed.

Results: As per GHQ-12 evaluation, after the deployment score increased from 0.2574 to 0.9162, but remained lower than the 'caseness' level of 2. Among the 79 troops with a score of 2 and more, the majority were married and had past history of tenures at high-altitude areas. There was statistically significant increase in the AFMC LES scores also on deployment.

Conclusion: Deployment at extremely high-altitude areas for even three months produces significant psychological morbidity among troops.

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Introduction

The deployment of Indian Armed Forces at the extreme High Altitude has been happening for more than thirty-five years. These are considered the toughest, coldest, highest and bravest battlefield in the world.[1]

Troops in forward posts are deployed at heights from 18,000 ft to 20,800 ft, temperatures dip below −55 °C and winds up to 160 knots are very common. More than the enemy

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shelling, troops have to fight the avalanches, hypoxia, isolation and boredom. As per a written reply to the Lok Sabha by the Minister of State for Defence Mr. Rao Inderjit Singh, a total of 869 troops have died in Operation Meghdoot until July 13, 2018, owing to climatic conditions, environmental and other factors.[2]

High-altitude stay impairs a number of cognitive performances, psychomotor performance, mental skills, memory and logical reasoning.[3] Studies have also shown enduring negative moods that approach the levels of clinical significance that may have implications on duty and performance of critical skills.[4] But most of the studies on effects of high-altitude areas (HAAs) are based on data from mountaineers, pilots, simulated hypoxia and artificial experimental situations. There are very few studies on the effects on psychological health of soldiers deployed and performing the duties at extreme HAAs.

Interestingly, there have been many studies on the effects of prolonged stay at Antarctica, which is a geographically remote, coldest, windiest and driest area with the highest average altitude on the planet, where photoperiods last as long as six months per year. Commonly reported psychiatric symptoms by people on polar expeditions include sleep disturbances (difficulty falling asleep and staying asleep), somatic symptoms (fatigue, aches), impaired cognitive abilities (memory, attention, reasoning), negative affect (depression, anger, anxiety) and interpersonal conflict and irritability.[5] In a study by Lawrence et al., incidence of diagnosable psychiatric disorders was 5.2% among personnel who spent an austral winter at two different research stations in Antarctica.[6] A systematic overview about the psychological effects on Antarctic expeditioners was carried out by Zimmer et al., assessing the published literature from 2000 to 2010.[7] They noted that frequent symptoms reported were cognitive impairment (63.6%), depression and low moods (56.8%), anxiety (47.7%) and irritability (45.4%). A reduced sense of general satisfaction was reported in 25% of the publications, particularly in relation to satisfaction obtained from the work, followed by diminished productivity, low morale and decreased well-being and optimism.

To the best of our knowledge, there have been no published studies on the psychological effects of deployment at the extreme high altitude on Indian soldiers so far (as per the search conducted in databases, PubMed Central, Ovid and Google using key words mentioned in the abstract). Hence, this study was undertaken to assess the impact of a tenure at the extreme high altitude on troops at the predeployment and postdeployment phase using validated screening tools.

Materials and methods

This longitudinal cohort study was conducted on troops that are to be deployed at the extreme high altitude during the second half of 2017. After institutional ethical committee clearance, informed consent was obtained, and demographic details were taken using a specially prepared psychosocial pro forma. Troops belonged to all arms and services, although the majority were from infantry and artillery. Participation was voluntary, and those with past history of head injury or any medical and psychiatry treatment history were excluded. All the troops who had completed a mandatory acclimatisation schedule and fulfilling inclusion/exclusion criteria were included in the study. A total of 334 personnel from all ranks participated in this study.

All participants were provided with the self-administered General Health Questionnaire-12 (GHQ-12)[8] (Hindi version) and Armed Forces Medical College Life Events Scale (AFMC LES)[9] (Hindi version). These scales were administered under the supervision of medical officers. The GHQ-12 is a psychometric screening tool that indicates a measure of psychological health, which was developed by Goldberg.[8] It has 12 items, each rated on a four-point scale, and the scoring was carried out in a bimodal way (0-0-1-1). A cut-off score of 2 or more is considered adequate to consider 'caseness'. The AFMC LES has 52 items and is specially designed to assess the stressful life events of service personnel. It has a Cronbach's alpha of 0.861, indicating high internal consistency, and it also has high criterion validity. In addition to the number of significant life events, the scale also gives the Life Change Unit (LCU) score. After a deployment for more than about three months at an altitude ranging from 15,500 ft to 19,500 ft, the troops were deinducted, and the same sample of troops were again assessed using the same questionnaires. The data collected were analysed using one-sample Kolmogorov–Smirnov test and Wilcoxon signed-rank test in SPSS version 22 software.

Results

A total of 334 army personnel participated in this study. Demographic details and rank distribution are shown in Table 1. The mean age of the participants was 30.30 years, the mean service was 6.51 years, the majority were married (59%) and most of them were educated up to the higher secondary level (88.9%).

There was increase in the scores of both screening tools, namely, GHQ-12 and AFMC LES, during the postdeployment phase compared with the predeployment phase, as depicted in Table 2. Seventy-nine troops achieved a GHQ-12 score of 2 or more, indicating 'caseness' level. The majority of them

Table 1 – Demographic profile of the study participants (N = 334).

Mean age in years (SD)	30.30 (6.51)
Mean years of service (SD)	10.88 (6.53)
Marital status	
Married	197 (59%)
Unmarried	137 (41%)
Education	
Matric	148 (44.3%)
Higher secondary	149 (44.6%)
Graduate	33 (9.88%)
Postgraduate	04 (1.19%)
Rank	
Officers	05 (1.49%)
JCOs	06 (1.79%)
OR	323 (96.72%)

JCOs = Junior Commissioned Officer Other Ranks; SD = standard deviation.

were married, aged between 20 and 30 years, were in the service bracket of more than 10 years and had history of past deployment in HAAs.

The scores of all troops in all the parameters were assigned a positive rank when the postdeployment score was more than the predeployment score. Similarly, they were assigned a negative rank when the postdeployment score was less than the predeployment score. It was found that difference in the GHQ-12 score, difference in life events and difference in the LCU score do not follow normal distribution (one-sample Kolmogorov–Smirnov test; p-value <0.05). Hence, change in these variables was analysed using the Wilcoxon signed-rank test, which showed significance on the two-tailed test, as shown in Table 3.

Table 3 – Change in psychometric scale scores after deployment (Wilcoxon signed-rank test).

Scale	Median (IQR)	Z	Wilcoxon signed-rank two-tailed test, p-value
GHQ-12	0 (1)	-8.544	0.000
Mean number o f life events	0 (1.25)	-7.336	0.000
Mean LCU score	0 (75)	–7.065	0.000

IQR = interquartile range; LCU = Life Change Unit; GHQ-12 = General Health Questionnaire-12.

Discussion

The adverse effects of serving at high altitudes on physical health of troops have been studied in detail, and the literature is replete with them. Studies on clinical conditions, such as heart rate variability,[10] symptomatic portal system thrombosis,[11] subacute mountain sickness,[12] gastric perforation[13] and pulmonary embolism,[14] in Indian soldiers serving at HAAs have also been conducted. The role of yoga practice in improving health and performance of soldiers serving at HAAs has also been reported.[15] However, no published research literature on the psychiatric morbidity of our soldiers deployed at extreme high altitude was available, although there were few studies from other countries.

A study on changes in the cognitive state of 129 Pakistani troops that were deployed at a mean altitude of more than 4500 m and that stayed there for a period of 111 days had revealed an overall decrease in their cognitive efficiency.[16] In another study, psychiatric morbidity of 245 Pakistani troops that 'acclimatised and deployed' for a minimum of seven weeks in high altitude and that returned to the base camp was compared with troops 'acclimatised but not deployed' using the GHQ-12[8] and Present State Examination[17] as screening tools.[18] In the deployed group, 56.3% had psychiatric morbidity, whereas the non-deployed group had only 30.2%. The prevalence of depression and anxiety symptoms was much higher in the deployed group, and psychotic symptoms were seen in the deployed group only.

Recently, a study was conducted on Pakistani troops that were deployed and serving at an altitude of 15,000 ft at Karakoram ranges; psychiatrists evaluated them on-site using the GHQ-12 (Urdu version), Hamilton Anxiety Depression Scale (HADS)[19] and clinical interview.[20] As per GHQ-12 evaluation, the 'caseness' level was seen in 74%, and as per HADS evaluation, 39% scored on depression and 47% scored on anxiety. Beghel et al.[21] had stated that Pakistani army had termed the psychological effect of fighting at high altitudes as 'Siachen syndrome', describing the progressive change in personality of the soldiers from normal to selfish, then introverted and finally irrational.

Zhao et al.[22] did a cross-temporal meta-analysis of Chinese military mental health at high altitudes. Thirty-eight research articles of research on mental health from 1993 to 2013 that used the Symptom Check-List-90[23] were analysed. They noted that over the previous twenty years, Chinese military mental health at high altitudes had experienced three stages of mental health, that is, a deteriorating, a fluctuating and then a stable period.

In this study, evaluation of psychological health of troops before and after deployment at the high altitude was undertaken. A total of 334 troops of all ranks participated in this study (Table 1). The mean GHQ-12 score at the predeployment phase was 0.2574, which increased to 0.9162 after more than three months of deployment (Table 2). Although this increase is statistically significant, it was much below the 'caseness' cut-off score of 2. However, 79 (23.7%) troops had scores of 'caseness' level after deployment. This is in contrast to the findings of the study by Ahmed et al.,[20] in which 'caseness' was seen in 74% of Pakistani troops deployed in a similar environment. In another study by Bashir[18] on Pakistani troops deployed in Siachen, 55.9% reached the 'caseness' level. No definite explanations can be offered for this finding of a low level of psychiatric morbidity among our troops, although this has been brought in the literature in the past also.[24] It can only be conjectured that possibly the psychosocial support by strong esprit de corps in such a harsh environment may be the biggest psychoprotective factor. However, a future study that includes a validated scale to measure social support within the Indian Army units would be able to elucidate and ascertain these protective psychosocial factors.

Of the 79 troops that achieved the 'caseness' level, 58 (73.4%) were married and 51 (65%) had history of past

Table 2 – Change in mean scores of parameters after deployment (N = 334).

Study parameter	Predeployment score	Postdeployment score	Difference
GHQ-12	0.2574	0.9162	0.6587 (+256%)
AFMC LES, number of life events	3.9251	4.6018	0.6766 (+17.24%)
AFMC LES, LCU score	171.9521	199.9192	27.9671 (+16.26%)

GHQ-12 = General Health Questionnaire-12; AFMC LES = Armed Forces Medical College Life Events Scale; LCU = Life Change Unit.

deployment in HAAs. Similar findings had been reported by Ahmed et al. also.[20] Pressing domestic issues and sense of inability to contact or to be available to family members when needed may be the reasons for higher morbidity among married soldiers. Similarly, knowledge of hardships and dangers of HAA tenures with past experience may be reasons for higher morbidity seen among experienced soldiers. No significant association of scores with years of service or educational status was noted.

The concept of life change units emphasises cumulative effects of stress, so importance is given to the number of life events witnessed by troops. The life-change model assumes that all changes in a person's life—large or small, desirable or undesirable—can act as stressors and that the accumulation of several small changes can be as powerful as one major change. Based on the stress potential, weightage in the form of LCU scores is assigned to each event of the scale. The fact is that the cumulative impact of many events can be considered to impose health hazards both at the physical and mental level. It is pertinent to focus on evaluating the impact of such life change units on psychological health of soldiers.

When the difference in scores from the postdeployment to predeployment phase was compared, they were statistically significant for GHQ-12, LES and LCU scores for the last one year. When the change in scores was assigned ranks for each person (positive when the postdeployment score was higher and negative when it was lower than the predeployment score), there were a very high number of positive rank scores in all the screening tools. As per the Wilcoxon signed-rank test, they were all found to be statistically significant (Table 3). The mean number of life events increased from 3.9251 to 4.6018 after deployment. However, it was just marginally more than the normative mean score of 4.18 for a soldier in field.[25]

On the other hand, the LCU score in the past one year increased from 171.95 to 199.92 after deployment. It was much higher than the normative mean score of 119.11 for a soldier in field. Analysis revealed this to be due to the presence of combination of high scoring life events such as fighting against terrorists (a score of 61), tenures in HAAs (a score of 51) and tenures in operational areas (a score of 50) in the last one year for many of the troops. This was probably due to their tenure in operational/counterinsurgency areas within the last one year before the deployment at the extreme high altitude. Such kinds of cumulative or prolonged tenures in operational areas continuously have the potential to have detrimental effects on the psychological well-being of soldiers, as already brought in by a study conducted on troops deployed in counterinsurgency areas for long.[26]

Conclusion

In this study, troops posted at the extreme high altitude were evaluated before and after deployment for their psychological health status. Within a short span of three months of deployment, there has been some deterioration in their psychological health status as measured by reliable and valid screening tools, although not reaching to the level of clinical morbidity. In this most inhospitable terrain of battlefield in the world, troops are likely to develop various psychological stresses, and hence, commanders at all levels are required to be acutely aware of this. It is also recommended to evaluate long-term effects of deployment in extreme HAAs in future. To the best of knowledge of the authors, this is the first study to evaluate the psychological health of Indian troops deployed at the extreme high altitude.

Disclosure of competing interest

The authors have none to declare.

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